Systems & Inequalities

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 16 cm. Multiplying the original length by the scale factor gives $4 \times 4 = 16$ cm.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 4)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.

Final Answer: 16. Substituting $x = 3$ into $y = 4x + 4$ gives $y = 4(3) + 4 = 16$.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In ΔABC studying geometric sequences as exponential functions, $m\angle A = 30^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 84°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (30^\circ + 66^\circ) = 84^\circ$.

4 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

For the model $f(x) = 3x + 5$ in comparing exponential relationships, evaluate $f(2)$.

Final Answer: 11. Substitute $x = 2$: $f(2) = 3(2) + 5 = 11$.

5 **[Foundational]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

A data set recorded for comparing exponential relationships consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 22.5. The sum is $16 + 20 + 24 + 30 = 90$, and dividing by 4 gives 22.5.

6 **[Foundational]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

For the model $f(x) = 4x + 6$ in comparing linear and exponential relationships, evaluate $f(2)$.

Final Answer: 14. Substitute $x = 2$: $f(2) = 4(2) + 6 = 14$.

7 **[Foundational]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

A data set recorded for comparing linear and exponential relationships consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 23.5. The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

© ClipSAT — clipsat.org | All Rights Reserved

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 5x + 7$ in simple interest, evaluate $f(2)$.

Final Answer: 17. Substitute $x = 2$: $f(2) = 5(2) + 7 = 17$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Solve the linear system using matrix inversion or Cramer's Rule: $3x - 2y = 8\ 5x + 4y = 6$

Write the system in matrix form $A\mathbf{x} = \mathbf{b}$: $(3, -2; 5, 4)(x; y) = (8; 6)$. Calculate the determinant of coefficient matrix A: $\det A = (3)(4) - (-2)(5) = 12 - (-10) = 22$. Find the inverse matrix A^{-1}: $A^{-1} = \frac{1}{22}(4, 2; -5, 3)$. Multiply $A^{-1}\mathbf{b}$.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Level 3 · AdvancedIn the system of equations below, k is a constant. For what value of k does the system have no solution? $\{kx - 6y = 15; 4x - 3y = 9$

For a system of two linear equations in standard form $Ax + By = C$ to have no solution, the lines must be parallel (equal ratios of x- and y-coefficients, but unequal ratio of constants): $\frac{k}{4} = \frac{-6}{-3} eq \frac{15}{9}\ \frac{k}{4} = 2 \Rightarrow k = 8$. Check constant ratio: $\frac{15}{9} = \frac{5}{3} eq 2$, confirming the lines never intersect. Thus, $k = 8$.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Level 4 · SynthesisIn the system of equations $ax + 8y = 24$ and $6x + by = 8$, a and b are constants. If the system has infinitely many solutions, what is the value of $a + b$?

The constant ratio from the first equation to the second is $\frac{24}{8} = 3$. For infinitely many solutions, every coefficient in the first equation must be 3 times the corresponding coefficient in the second: $a = 3 \times 6 = 18, \quad 8 = 3b \Rightarrow b = \frac{8}{3}.\ a + b = 18 + \frac{8}{3} = \frac{62}{3}$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 2 · MediumA theater sold a total of 250 adult and student tickets for an evening performance, collecting $2,400. If adult tickets cost 12 each and student tickets cost 8 each, how many student tickets were sold?

Let a be adult tickets and s be student tickets: $\{a + s = 250; 12a + 8s = 2400$ Multiply the first equation by 12: $12a + 12s = 3000$. Subtract the second equation: $(12a + 12s) - (12a + 8s) = 3000 - 2400 \Rightarrow 4s = 600 \Rightarrow s = 150$. The theater sold 150 student tickets.

© ClipSAT — clipsat.org | All Rights Reserved

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 1 · FoundationsIf $5(2x-3)=3(x+4)+6$, what is the value of $7x-15$?

Expand both sides: $10x-15=3x+12+6 \Rightarrow 10x-15=3x+18$. $7x=33 \Rightarrow x=\frac{33}{7}$. Substitute into the target expression $7x-15$: $7\left(\frac{33}{7}\right)-15=33-15=18$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Matrices & Systems · Practice 3] If the governing relation for this system is $g(t)=5t+5$, evaluate $g(8)$ and find the inverse input t when $g(t)=50$.

Substitute $t=8$: $g(8)=5(8)+5=\mathbf{45}$. Setting $5t+5=50$ gives $5t=45 \Rightarrow t=\mathbf{9}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 10.01 Rational Expressions

Find the value(s) of k for which $A=(k, 2; 3, k)$ is singular.

$A=k^2-6=0k=6$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 11.01 Statistics & Data Analysis

The equation of a circle in the xy-plane is $(x-5)^2+(y+2)^2=81$. What is the radius of the circle?

In standard circle form $(x-h)^2+(y-k)^2=r^2$, the constant on the right side represents r^2. Thus $r^2=81 \Rightarrow r=9$. Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Quadratic Functions

A solid metal cone with base radius 4 cm and height 9 cm is submerged into a cylindrical beaker with internal radius 6 cm partially filled with water. Assuming the cone is fully submerged and no water overflows, by how many centimeters does the water level in the beaker rise?

Step 1: Compute the volume of the submerged cone: $V_{\text{cone}}=\frac{1}{3}\pi r_{\text{cone}}^2 h_{\text{cone}}=\frac{1}{3}\pi(4^2)(9)=\frac{1}{3}\pi(16)(9)=48\pi$ cm^3. Step 2: The volume of water displaced forms a cylinder with radius $R=6$ cm and height Δh: $V_{\text{displaced}}=\pi R^2(\Delta h)=\pi(6^2)\Delta h=36\pi\Delta h$. Step 3: Equate volumes and solve for Δh.

© ClipSAT — clipsat.org | All Rights Reserved

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Exponents & Polynomials

Solve the rational equation for x: $\frac{x}{x-2} + \frac{1}{x+3} = \frac{5}{x^2+x-6}$. Check for extraneous solutions.

Factor the quadratic denominator: $x^2 + x - 6 = (x-2)(x+3)$. The LCD is $(x-2)(x+3)$, with restricted domain $xeq2, -3$. Multiply both sides by $(x-2)(x+3)$: $x(x+3) + 1(x-2) = 5$.
$x^2 + 3x + x - 2 = 5 \Rightarrow x^2 + 4x - 7 = 0$. Apply the quadratic formula:
$x = \frac{-4 \pm \sqrt{16 - 4(1)(-7)}}{2} = \frac{-4 \pm \sqrt{16+28}}{2} = \frac{-4 \pm \sqrt{44}}{2} = \frac{-4 \pm 2\sqrt{11}}{2} = -2 \pm \sqrt{11}$. Check restrictions: $-2 \pm \sqrt{11} \approx 1.317$ and -5.317. Neither equals.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Functions & Transformations

A committee of 4 members is to be selected at random from a club of 7 seniors and 5 juniors. (a) What is the total number of possible committees? (b) Find the probability that the committee contains exactly 2 seniors and 2 juniors.

(a) Order does not matter, so the total number of ways to choose 4 members from 12 total people is: $\text{Total} = \binom{12}{4} = \frac{12 \times 11 \times 10 \times 9}{4 \times 3 \times 2 \times 1} = 495$. (b) Number of ways to choose 2 seniors from 7: $\binom{7}{2} = \frac{7 \times 6}{2 \times 1} = 21$. Number of ways to choose 2 juniors from 5: $\binom{5}{2} = \frac{5 \times 4}{2 \times 1} = 10$. By the fundamental counting principle, favorable outcomes: $21 \times 10 = 210$. Probability.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Find all real solutions to the equation $|2x - 7| = 3x - 1$.

Step 1: Split into two cases based on the definition of absolute value: Case 1:
$2x - 7 = 3x - 1 \Rightarrow -7 + 1 = 3x - 2x \Rightarrow x = -6$. Check Case 1: $|2(-6) - 7| = |-19| = 19$, but the right side is $3(-6) - 1 = -19$. Since $19eq-19$, $x = -6$ is an extraneous solution. Case 2:
$2x - 7 = -(3x - 1) = -3x + 1 \Rightarrow 2x + 3x = 1 + 7 \Rightarrow 5x = 8 \Rightarrow x = \frac{8}{5} = 1.6$. Check Case 2: $|2(1.6) - 7| = |3.2 - 7| = |-3.8| = 3.8$. Right side: $3(1.6) - 1 = 4.8 - 1 = 3.8$ (Valid ✓). The o.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 5x + 9$ in graphs and characteristics of exponential functions, evaluate $f(2)$.

a. 19 ✓

b. 24

c. 10

d. 23

Substitute $x = 2$: $f(2) = 5(2) + 9 = 19$.

© ClipSAT — clipsat.org | All Rights Reserved

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 6x + 3$ in exponential growth and decay, evaluate $f(2)$.

a. 15 ✓

b. 21

c. 12

d. 19

Substitute $x = 2$: $f(2) = 6(2) + 3 = 15$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for graphs and characteristics of exponential functions consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for exponential growth and decay consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for simple interest consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5 ✓

d. 26.5

The sum is $18 + 22 + 26 + 32 = 98$, and dividing by 4 gives 24.5.

© ClipSAT — clipsat.org | All Rights Reserved

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For eigenvalue λ with eigenvector v, any scalar multiple cv is also an eigenvector. True or False?

a. Only if $c = 1$

b. False

c. Only if $c > 0$

d. **True ✓**

$A(cv) = c(Av) = c\lambda v = \lambda(cv)$. ✓.

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the xy-plane, what is the area of the triangular region bounded by the lines $y = \frac{1}{2}x$, $y = -2x + 10$, and the x-axis?

a. 4

b. **5 ✓**

c. 6

d. 10

The line $y = \frac{1}{2}x$ crosses the x-axis at $(0, 0)$. The line $y = -2x + 10$ crosses the x-axis at $(5, 0)$. The base along the x-axis has length $5 - 0 = 5$. Finding the intersection: $\frac{1}{2}x = -2x + 10 \Rightarrow 2.5x = 10 \Rightarrow x = 4$, so $y = 2$. The height is 2. Area $= \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 5 \times 2 = 5$.

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Line k passes through the points $(7, 2)$ and $(-3, 22)$. Line p is perpendicular to line k and passes through $(4, -3)$. What is the y-intercept of line p?

a. **-5 ✓**

b. 5

c. $-\frac{7}{2}$

d. -2

The slope of line k is $m_1 = \frac{22-2}{-3-7} = \frac{20}{-10} = -2$. Because line p is perpendicular to line k, its slope is the negative reciprocal: $m_2 = \frac{1}{2}$. Using point-slope form with $(4, -3)$:
$y - (-3) = \frac{1}{2}(x - 4) \Rightarrow y + 3 = \frac{1}{2}x - 2 \Rightarrow y = \frac{1}{2}x - 5$. Therefore, the y-intercept of line p is -5.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

© ClipSAT — clipsat.org | All Rights Reserved

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The graph shows the volume of water $V(t)$, in thousands of gallons, remaining in a municipal reservoir t hours after draining begins at a constant rate. According to the linear model, after how many hours will the reservoir be completely empty?

a. 8

b. **10** ✓

c. 12

d. 15

The constant rate of draining is $m = \frac{15-24}{5-2} = \frac{-9}{3} = -3$ thousand gallons per hour. Using point-slope form with $(2, 24)$: $V(t) - 24 = -3(t-2) \Rightarrow V(t) = -3t + 30$. The reservoir is completely empty when $V(t) = 0$: $0 = -3t + 30 \Rightarrow 3t = 30 \Rightarrow t = 10$ hours.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The line $y = kx + 7$ intersects the line $y = -2x + 1$ at a point with a positive x-coordinate and a positive y-coordinate. Which condition must k satisfy?

a. $k > -2$

b. $k < -14$ ✓

c. $k > -7$

d. $k < -2$

Set $kx + 7 = -2x + 1 \Rightarrow (k+2)x = -6 \Rightarrow x = -\frac{6}{k+2}$. For $x > 0$, we must have $k + 20$ when $k + 2 < 0$, the numerator $k + 14$ must also be negative: $k + 14 < 0 \Rightarrow k < -14$.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Line k passes through $(-3, 7)$ and $(5, -9)$. What is the x-intercept of line k?

a. $(0.5, 0)$ ✓

b. $(1, 0)$

c. $(0, 1)$

d. $(2, 0)$

Slope $m = \frac{-9-7}{5-(-3)} = \frac{-16}{8} = -2$. Equation: $y - 7 = -2(x+3) \Rightarrow y = -2x + 1$. For x-intercept, set $y = 0$: $0 = -2x + 1 \Rightarrow 2x = 1 \Rightarrow x = 0.5$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

What is the distance between the parallel lines $y = 2x + 4$ and $y = 2x - 6$?

a. $2\sqrt{5}$ ✓

b. $\sqrt{10}$

c. 5

d. $2\sqrt{10}$

The distance between parallel lines $y = mx + b_1$ and $y = mx + b_2$ is $d = \frac{|b_1 - b_2|}{\sqrt{1+m^2}} = \frac{|4-(-6)|}{\sqrt{1+2^2}} = \frac{10}{\sqrt{5}} = 2\sqrt{5}$.

© ClipSAT — clipsat.org | All Rights Reserved

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For what value of b does the line $y = 3x + b$ pass through the intersection point of $y = 2x + 1$ and $y = -x + 10$?

a. -4

b. **-2** ✓

c. 2

d. 4

Find intersection: $2x + 1 = -x + 10 \Rightarrow 3x = 9 \Rightarrow x = 3$. Then $y = 2(3) + 1 = 7$. Substitute $(3, 7)$ into $y = 3x + b \Rightarrow 7 = 3(3) + b \Rightarrow b = -2$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A linear function has $g(0) = -3$ and its graph is perpendicular to the line $y = 4x + 7$. What is $g(12)$?

a. **-6** ✓

b. -4

c. 0

d. 3

Slope of the perpendicular line is $m = -1/4$. Since $g(0) = -3$, $g(x) = -\frac{1}{4}x - 3$. At $x = 12$, $g(12) = -\frac{1}{4}(12) - 3 = -3 - 3 = -6$.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Which values of x satisfy $-2x + 3 > 7$?

a. $x > -2$

b. $x < -2$ ✓

c. $x > 2$

d. $x < 2$

Applying Linear Functions & Linear Inequalities principles, $-2x > 4 \Rightarrow x < -2$ Therefore, the verified correct option is $x < -2$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A line passes through the points $(1, 12)$ and $(3, 16)$. What is its x-intercept?

a. 3

b. -11

c. -12

d. **-5** ✓

Slope $= 2, y-\text{intercept} = 10$. Setting $y = 0$: $0 = 2x + 10 \Rightarrow x = -5$.

© ClipSAT — clipsat.org | All Rights Reserved

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Which value of x satisfies the inequality $-3x + 8 < -7$?

a. 3

b. 4

c. 5

d. **6** ✓

Subtract 8: $-3x5$. Among the choices, only 6 is strictly greater than 5.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Exponents & Polynomials

The graph of a function is shown. At which x-value does the function equal zero?

a. $x = -2$

b. $x = 0$

c. $x = 1$

d. $x = 2$ ✓

The zero (x-intercept) is where the graph crosses the x-axis: $x = 2$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Functions & Transformations

Universal set $\mathscr{E} = \{x : x \in \mathbb{Z}^+, x \leq 15\}$. Set $A = \{x : x \text{ is a multiple of } 3\}$ and $B = \{x : x \text{ is a multiple of } 5\}$. Find $A \cap B$.

a. $\{15\}$ ✓

b. $\{3, 5, 15\}$

c. $\{3, 5, 6, 9, 10, 12, 15\}$

d. $\varnothing$

Multiples of both 3 and 5 are multiples of 15: only $\{15\}$. [B1]

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

A line passes through the points $(-1, 15)$ and $(3, 7)$. If $f(x)$ represents this line, what is $f(-6)$?

a. 29

b. 16

c. **25** ✓

d. 18

Slope $= \frac{7-15}{3+1} = -2$. $f(x) = -2x + 13$. $f(-6) = 25$.

© ClipSAT — clipsat.org | All Rights Reserved

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Sequences & Series

What is the slope of the line passing through $(2, 5)$ and $(6, 17)$?

a. 2

b. **3** ✓

c. 4

d. 5

$m = \frac{17-5}{6-2} = \frac{12}{4} = 3$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 8.01 Factoring Polynomials

Which value of x satisfies $3x - 7 > 11$?

a. 4

b. 5

c. 6

d. **7** ✓

$3x > 18 \Rightarrow x > 6$. Only $x = 7$ works.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 9.01 Radicals & Exponents

Consider the matrix equation $(2, 1, -2; 4, -1, -1; 2, -2, 1)(x; y; z) = (9; 9; k)$ where k is a real constant. Given that the system has infinitely many real solutions (x, y, z), what is the minimum possible value of $x^2 + y^2 + z^2$?

a. **9** ✓

b. 3

c. 6

d. 45

We solve the problem in three steps: determine k for consistency, parameterize the infinite solution set, and minimize the quadratic objective function. **Step 1: Determine k for consistency** Write the augmented matrix of the system: $(2, 1, -2, 9; 4, -1, -1, 9; 2, -2, 1, k)$ Perform Gaussian elimination: - Replace Row 2 with $R_2 - 2R_1$: $(4-4, -1-2, -1-(-4) \mid 9-18) = (0, -3, 3 \mid -9)$, which simplifies to $(0, 1, -1 \mid 3)$. - Replace Row 3 with $R_3 - R_1$.

© ClipSAT — clipsat.org | All Rights Reserved

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Linear Equations & Lines

Consider the system of linear equations represented by the matrix equation $(1, 1, 1; 1, 2, 3; 3, 5, k)(x; y; z) = (6; 14; m)$ where k and m are real constants. Suppose the system is consistent and has infinitely many solutions. Let $P(x, y, z)$ be the unique solution that minimizes the squared Euclidean distance from the origin, $d^2 = x^2 + y^2 + z^2$. What is the value of $k + m + d^2$?

a. **55 ✓**

b. 47

c. 38

d. 109

We solve the problem in three main stages: 1. **Determine the parameter k:** For the system to have infinitely many solutions, the coefficient matrix must be singular (its determinant must be zero): $\det(1, 1, 1; 1, 2, 3; 3, 5, k) = 1(2k - 15) - 1(k - 9) + 1(5 - 6) = k - 7$ Setting $k - 7 = 0$ yields $k = 7$. 2. **Determine the parameter m:** Set up the augmented matrix with $k = 7$ and perform Gaussian elimination.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 10.01 Rational Expressions

A local government is planning to build a new community park. The park will feature a rectangular garden area and a circular fountain. The garden's length, L, and width, W, are constrained by a perimeter of at most 200 meters. Specifically, $2L + 2W \leq 200$. Additionally, to ensure accessibility, the width must be at least 10 meters less than the length, so $W \geq L - 10$. The area of the garden is given by $A = LW$. The city wants to maximize the garden's area, A, subject to these constraints. If the length of the garden, L, is restricted to be between 30 and 70 meters inclusive, what is the maximum possible area of the garden?

a. 2100 square meters

b. 2400 square meters

c. **2500 square meters ✓**

d. 2600 square meters

The constraints are: 1. $2L + 2W \leq 200 \Rightarrow L + W \leq 100$ 2. $W \geq L - 10$ 3. $30 \leq L \leq 70$ We want to maximize $A = LW$. From constraint 1, $W \leq 100 - L$. Combining this with constraint 2, we have $L - 10 \leq W \leq 100 - L$. For a valid range of W, we must have $L - 10 \leq 100 - L$, which means $2L \leq 110 \Rightarrow L \leq 55$. Thus, the effective range for L is $30 \leq L \leq 55$. To maximize $A = L(100 - L)$ (assuming W takes its maximum possible value from constraint 1 to maxim.

© ClipSAT — clipsat.org | All Rights Reserved